FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

2

RULE – BASED AGENT

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I. BACKGROUND INFORMATION

A Rule-Based Agent is a type of intelligent agent that operates by following predefined rules to decide its actions. In this lab, students will work with a vacuum cleaner agent that can clean rooms or move between rooms depending on their state (dirty or clean).

The goal is to understand how agents perceive the environment, act according to rules, and update the environment.

II. TECHNICAL ASSESSMENT

RULE – BASED AGENT

Objective

At the end of this laboratory activity, students are expected to:

1. Explain the concept of rule-based agents.
2. Implement a rule-based agent in Python.
3. Interact with an environment by perceiving and acting based on rules.
4. Observe and analyze the behavior of an agent in a controlled environment.

Materials Needed

- Computer with Python 3.x installed
- Text editor or IDE (e.g., VSCode, PyCharm)
- Optional: Jupyter Notebook for interactive testing

General Instructions

1. This **Technical Assessment (TA4)** must be completed individually.
2. Follow the Task below.
3. Include clear comments in your code to describe its logic.
4. Each program should display the correct input, process, and output.
5. Ensure that your script displays clear and readable outputs.
6. Compile all Python codes and corresponding outputs into one PDF file.
7. Save it as **TA2_lastname.pdf**
   (Example: TA2_Patilano.pdf)
8. Submit the **python file** and **PDF** file.

Laboratory Task

Pre-Lab Preparation

1. Review the Python concepts:
   - Classes and objects
   - If-Else statements

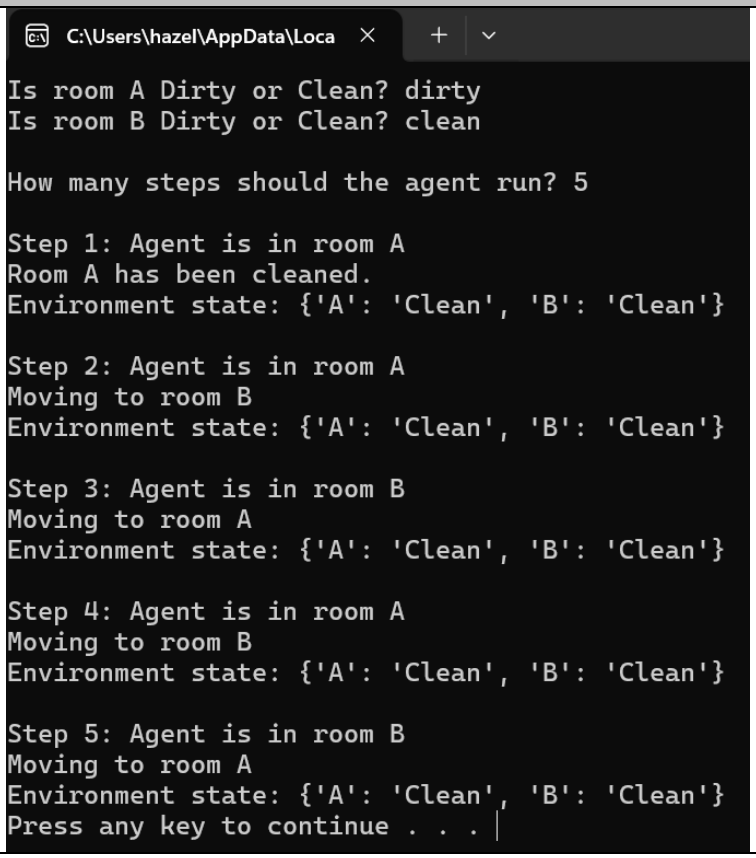
- Loops (for loop)
- Functions / methods

2. Understand the environment:
   - Two rooms: A and B
   - Each room can be Dirty or Clean

Task 1: Set Up the Environment

1. Initialize two rooms using a Python dictionary.
2. Allow the user to set the **initial state** of each room (dirty or clean).
3. Write a function to **check if a room is dirty** and a function to **clean a room**.

Sample Output



Task 2: Implement the Rule-Based Agent

1. Create a class for the agent.
2. Implement the **perceive-and-act method**:
   - Rule 1: If the current room is dirty, clean it.
   - Rule 2: If the current room is clean, move to the other room.
3. Implement a **move method** to switch rooms.

Task 3: Run the Simulation

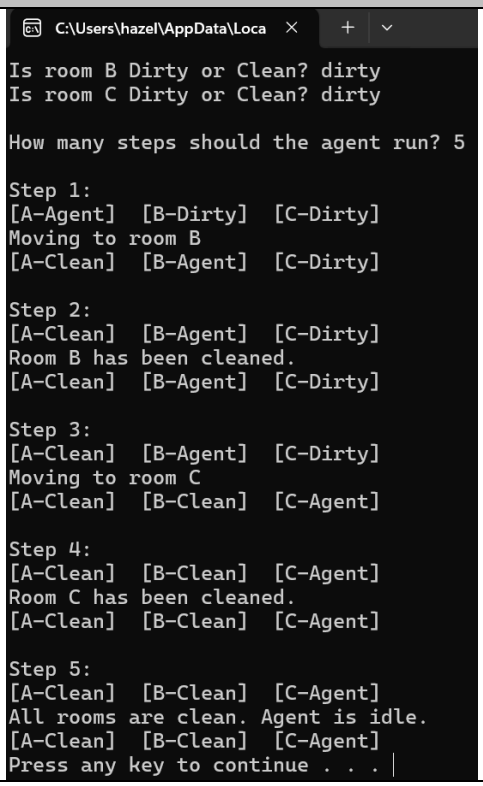
1. Ask the user how many steps the simulation should run.
2. Run the agent in a loop for the specified steps.

3. After each step, print:
    - The agent's current location
    - The action taken (cleaned/moved)
    - Current state of the environment

Optional (Bonus Task)

- Add a **third room (C)** and update the agent logic.
- Make the agent **randomly choose which dirty room to clean next**.
- Visualize the agent moving between rooms using a **text-based grid**.

Sample Output (Bonus Task)



Your Code - Laboratory Tasks ⇩

Source code:

```
#Task 1
#initializing two rooms using python dictionary

rooms = {
    "Room A": input("Is Room A Dirty or clean?: "),
    "Room B": input("Is Room B Dirty or clean?: ")
```

```
}

#creating function to check if the room is dirty.
def is_dirty(room):
    return rooms[room] == "dirty"

#creating a function to clean a room.
def clean_room(room):
    rooms[room] = "clean"
#Task 2
#creating class for agent
class Agents:
    #set the agent's starting location
    def __init__(self, location):
        self.location = location

    #decide what action to take based on the room's state using
perceive and act method
    def perceive_and_act(self):

        #Rule 1: If the current room is dirty clean it.
        if is_dirty(self.location):
            clean_room(self.location)
            return "Cleaned"
        #Rule 2: If the current room is clean, move to the other
room
        else:
            self.move()
            return "Moved"
    #implement move method to move agent to the other room
    def move(self):
        if self.location == "Room A":
            self.location = "Room B"
        else:
            self.location = "Room A"

#Task 3
 #Asking the user where the agent start.
start = input("Enter the starting room (A/B): ").upper()
if start == "A":
    start = "Room A"
else:
    start = "Room B"

#create the agent
```

```
agent = Agents (start)
steps = int(input("Enter number of steps: "))

#Run the agent for specified number of steps.
for step in range (1, steps +1):

 #let the agent perceive the room and perform an action.
    action = agent.perceive_and_act()

    #displaying results

    print("\nStep", step)
    print ("Location:", agent.location)
    print ("Action:", action)
    print ("Rooms:", rooms)
```

Output:

```
PS C:\Users\Dom> python -u "d:\Feu Alabangseu SY 26 - 27
Is Room A Dirty or clean?: dirty
Is Room B Dirty or clean?: clean
Enter the starting room (A/B): A
Enter number of steps: 5

Step 1
Location: Room A
Action: Cleaned
Rooms: {'Room A': 'clean', 'Room B': 'clean'}

Step 2
Location: Room B
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}

Step 3
Location: Room A
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}

Step 4
Location: Room B
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}

Step 5
Location: Room A
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}
PS C:\Users\Dom>
```

```
PS C:\Users\Dom> python -u "d:\Feu Alabangseu SY 2
Is Room A Dirty or clean?: clean
Is Room B Dirty or clean?: clean
Enter the starting room (A/B): b
Enter number of steps: 5

Step 1
Location: Room A
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}

Step 2
Location: Room B
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}

Step 3
Location: Room A
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}

Step 4
Location: Room B
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}

Step 5
Location: Room A
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}
PS C:\Users\Dom>
```

```
PS C:\Users\Dom> python -u "d:\Feu Alabangseu SY 26 - 27
Is Room A Dirty or clean?: clean
Is Room B Dirty or clean?: dirty
Enter the starting room (A/B): a
Enter number of steps: 5

Step 1
Location: Room B
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'dirty'}

Step 2
Location: Room B
Action: Cleaned
Rooms: {'Room A': 'clean', 'Room B': 'clean'}

Step 3
Location: Room A
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}

Step 4
Location: Room B
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}

Step 5
Location: Room A
Action: Moved
Rooms: {'Room A': 'clean', 'Room B': 'clean'}
PS C:\Users\Dom>
```

Task 4: Analysis Questions

After running the simulation, answer the following:

1. What rules does the agent follow?
2. How does the agent decide when to clean and when to move?
3. What happens if both rooms are already clean at the start?
4. How would you modify the agent to handle **more than two rooms**?

Answer the Questions ⇩

1. Based on case scenario the agent has two rules to follow: If the current room is dirty, it cleans the room otherwise it moves to the other room if the room is cleaned.

2. The agent checks if the current room is clean or dirty. If the room is dirty the agent cleans it. If it is clean, the agent moves to the other room.

3. If both rooms are already clean at the start, the agent will move from one room to another. Since according to the rule if the room is cleaned, the agent will move to another room.

4. To handle and modify the agent to handle more than two rooms, we can do it by adding more rooms to the dictionary and update the movement logic allowing it to move from one room to the next.